

Word Embeddings — Cheat Sheet (■■■■ 10)

Definition: Word embeddings map words to vectors in $\mathbb{R}^d$ preserving semantic similarity.

Cosine Similarity:

$$\cos(\theta) = (\mathbf{v}_1 \cdot \mathbf{v}_2) / (\|\mathbf{v}_1\| \cdot \|\mathbf{v}_2\|)$$

Main Algorithms:

- Word2Vec: CBOW, Skip-gram
- FastText (subword n-grams)
- GloVe

Skip-gram Softmax:

$$P(\mathbf{w}_o | \mathbf{w}_i) = \exp(\mathbf{v}_{\mathbf{w}_o}^T \mathbf{v}_{\mathbf{w}_i}) / \sum \exp(\mathbf{v}_{\mathbf{w}}^T \mathbf{v}_{\mathbf{w}_i})$$

Negative Sampling Loss:

$$\log \sigma(\mathbf{v}_{\mathbf{w}_o}^T \mathbf{v}_{\mathbf{w}_i}) + \sum \log \sigma(-\mathbf{v}_{\mathbf{w}_k}^T \mathbf{v}_{\mathbf{w}_i})$$

Document Embeddings:

Doc2Vec (PV-DM, PV-DBOW)

Visualization:

- PCA
- t-SNE
- UMAP

Libraries:

- gensim (Word2Vec, FastText, Doc2Vec)
- numpy
- scikit-learn
- matplotlib
- torch.nn.Embedding

Use Cases:

Classification, Sentiment Analysis, Semantic Search, Recommendation Systems